

v5.3

Mass resolution and observed upper limits

This study starts each dataset at its unscaled Monte Carlo (MC) mass resolution and follows four steps toward the documented scaled resolution. For the full 2015 and 2016 samples and the native 2021 10% sample, it compares individual observed limits from Gaussian-process (GP) background fits, the available-dataset combination, and fixed-mass asymptotic p-values.

The main result is a set of exploratory refitted resolution scenarios. The upward change from the MC curve to the largest of the four limits is a useful diagnostic envelope. It is not a propagated calibration uncertainty or a newly calibrated confidence interval.

Dataset	Mass range [MeV]	Nodes	Scaled / MC limit median [min, max]
2015	19-90	72	1.29 [0.771, 2.77]
2016	39-180	142	1.8 [0.42, 7.76]
2021	50-250	201	1.27 [1.1, 1.95]
Combination	19-250	232	1.31 [0.733, 2.77]

Ratios are point-by-point comparisons on the shared mass grid; their median is a descriptive summary over scanned masses. A ratio above one means that the fully scaled resolution gives a weaker observed upper limit than the MC baseline.

Narrower resolution and fitted uncertainty. The median ratios of the fully scaled to MC fitted signal-yield error are 2015: 1.33; 2016: 1.8; 2021: 1.26. These diagnose the change in total fitted error, including the background constraint; they do not isolate the GP contribution.

No expected-limit bands or toy ensembles were generated for this bounded study. All p-values are local, conditional asymptotic references. The study does not replace the frozen production result.

The four resolution scenarios

For each dataset d , the width at mass m is interpolated between its source MC and scaled curves: $\sigma_{d,t}(m) = \sigma_{d,MC}(m) + t [\sigma_{d,scaled}(m) - \sigma_{d,MC}(m)]$, with $t = 0, 0.4, 0.6$ and 1 . For a constant 2021 endpoint ratio of 1.25, this gives width multipliers 1.00, 1.10, 1.15 and 1.25. A mass-dependent correction remains mass dependent.

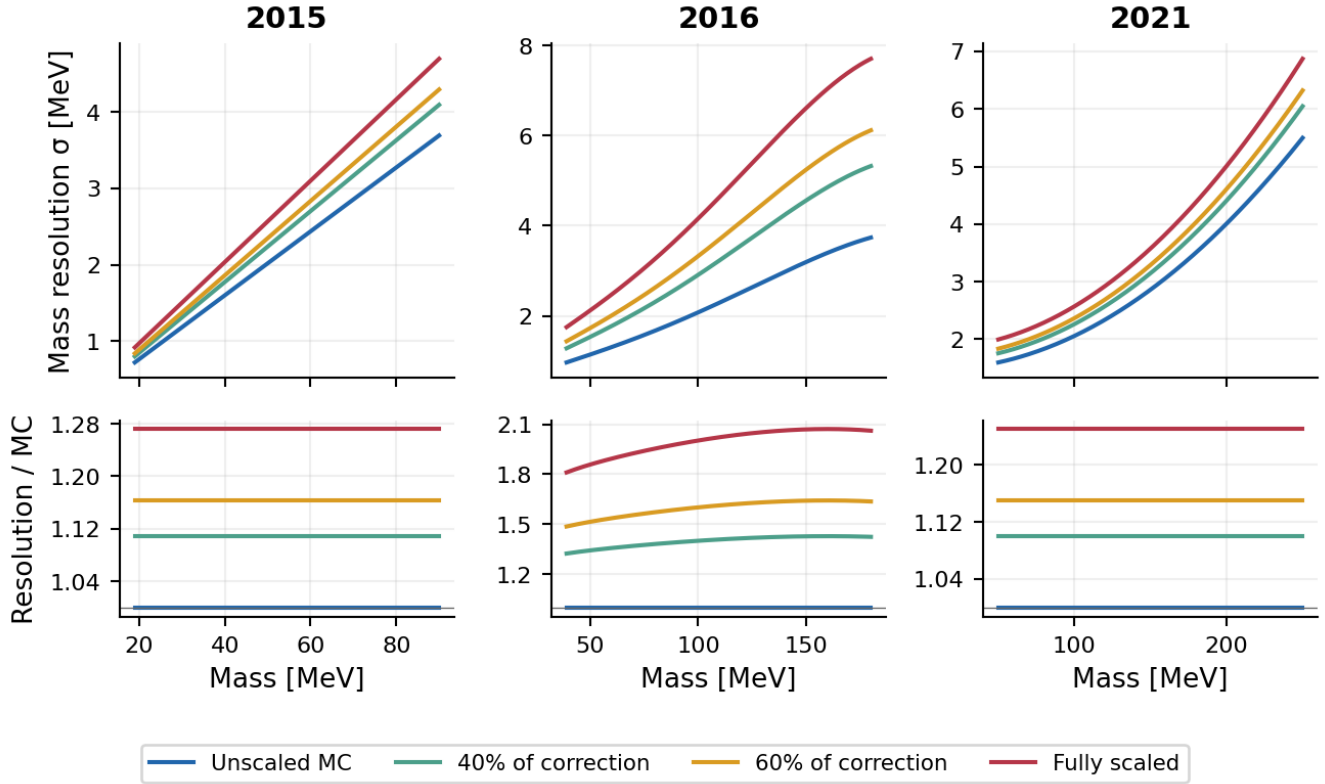


Figure 1. Signal mass resolutions and their ratios to the MC curve. Colors are used consistently in all subsequent plots. The 2015 multipliers are 1.000, 1.109, 1.163 and 1.272. The 2016 intermediate ratios vary with mass; the source polynomials and tail treatment are recorded in the protocol and resolution audit.

Each main scenario reruns the background procedure on the corresponding width-dependent window and signal exclusion. GP hyperparameters use 12 seeded restarts plus an admissible reviewed-kernel seed and its local optimization. Selection uses only the sideband marginal likelihood, uniformly at every coordinate; it does not guarantee the global optimum. The signal template and prompt-density conversion also change.

The limit fit uses a Poisson count likelihood with the saved GP background covariance as a Gaussian constraint and a shared nonnegative signal strength for the combination. The observed and background-Asimov likelihood ratios are profiled; their sampling tails use asymptotic CL_s . The upper limit solves $CL_s = 0.10$. The fixed-mass discovery reference uses $p_0 = 1 - \Phi(\sqrt{q_0})$, with $p_0 = 0.5$ at the physical boundary.

The plotted ϵ^2 includes the inherited dimuon branching correction above threshold, applied identically to every scenario and the released reference. It cancels in the resolution ratios and does not change p_0 . Raw electron-channel values remain in the scan ledger.

2015: unscaled and scaled observations

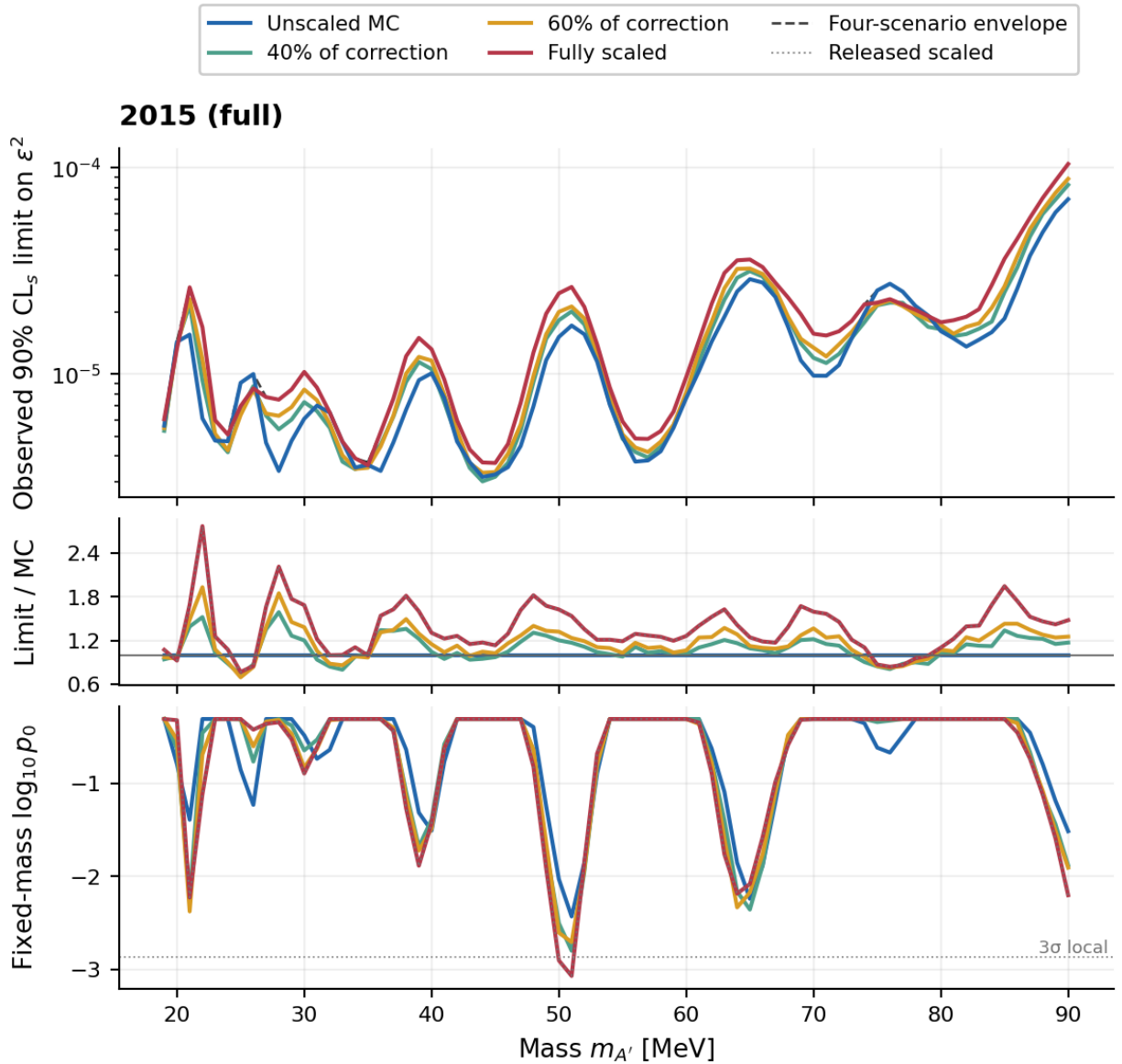


Figure 2. The upper panel compares observed 90% CL_s limits; the middle panel divides each curve by its MC baseline; the lower panel shows fixed-mass $\log_{10} p_0$. The dashed curve is the four-scenario maximum, without a probability interpretation. The gray dotted curve is the frozen scaled release, whose numerical comparison is discussed on page 9.

The fully scaled / MC limit ratio has median 1.29 and spans 0.771-2.77. The maximum four-scenario envelope / MC ratio is 2.77. An intermediate width supplies the largest limit at 1 of 72 masses with all four scenarios.

The smallest MC-scenario p_0 is 0.00371 at 51 MeV; the fully scaled minimum is 0.000847 at 51 MeV. These minima may occur at different masses and are not global p-values.

2016: unscaled and scaled observations

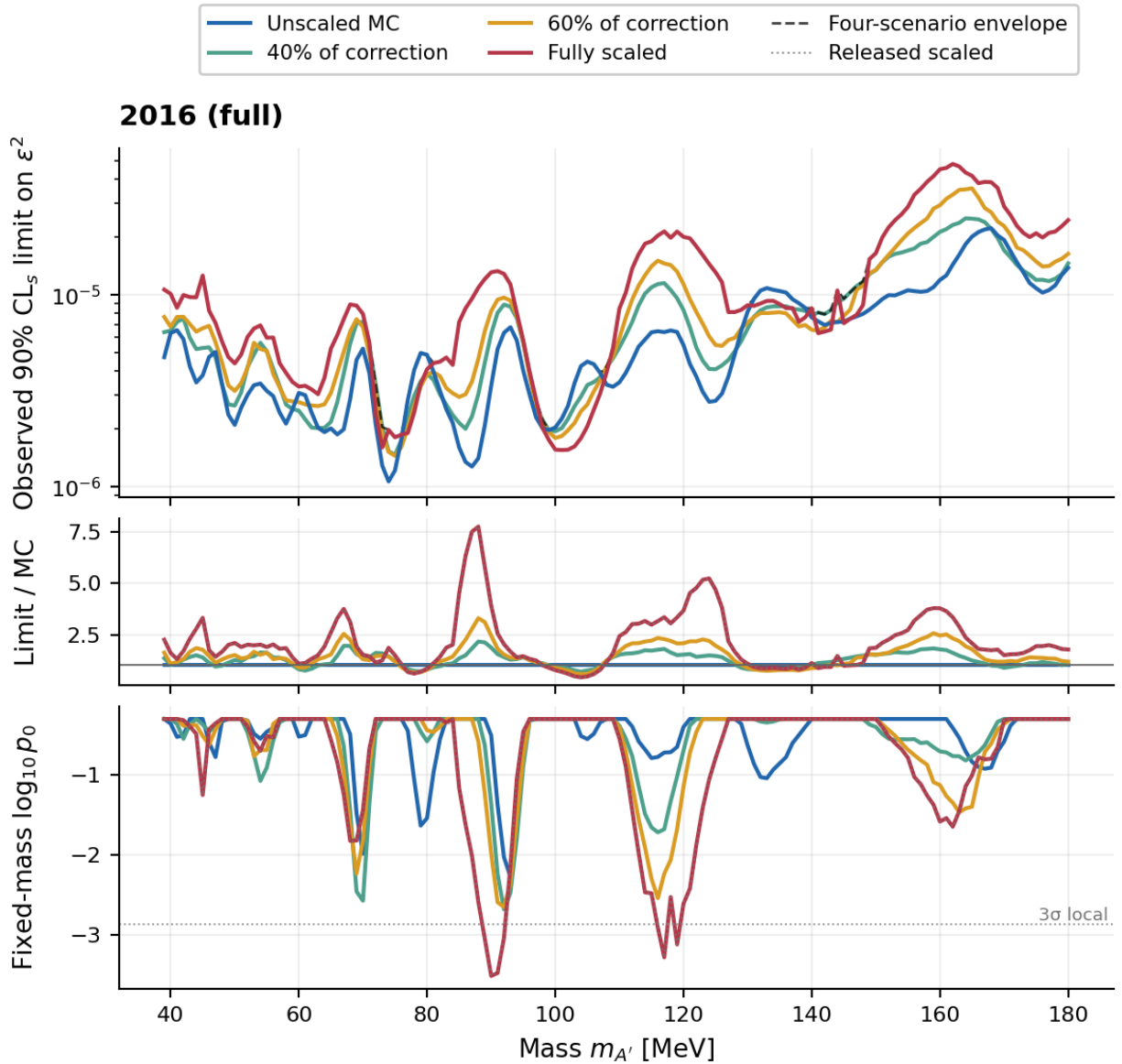


Figure 3. The upper panel compares observed 90% CL_s limits; the middle panel divides each curve by its MC baseline; the lower panel shows fixed-mass $\log_{10} p_0$. The dashed curve is the four-scenario maximum, without a probability interpretation. The gray dotted curve is the frozen scaled release, whose numerical comparison is discussed on page 9.

The fully scaled / MC limit ratio has median 1.8 and spans 0.42-7.76. The maximum four-scenario envelope / MC ratio is 7.76. An intermediate width supplies the largest limit at 11 of 142 masses with all four scenarios.

The smallest MC-scenario p_0 is 0.00533 at 93 MeV; the fully scaled minimum is 0.000308 at 90 MeV. These minima may occur at different masses and are not global p-values.

2021: unscaled and scaled observations

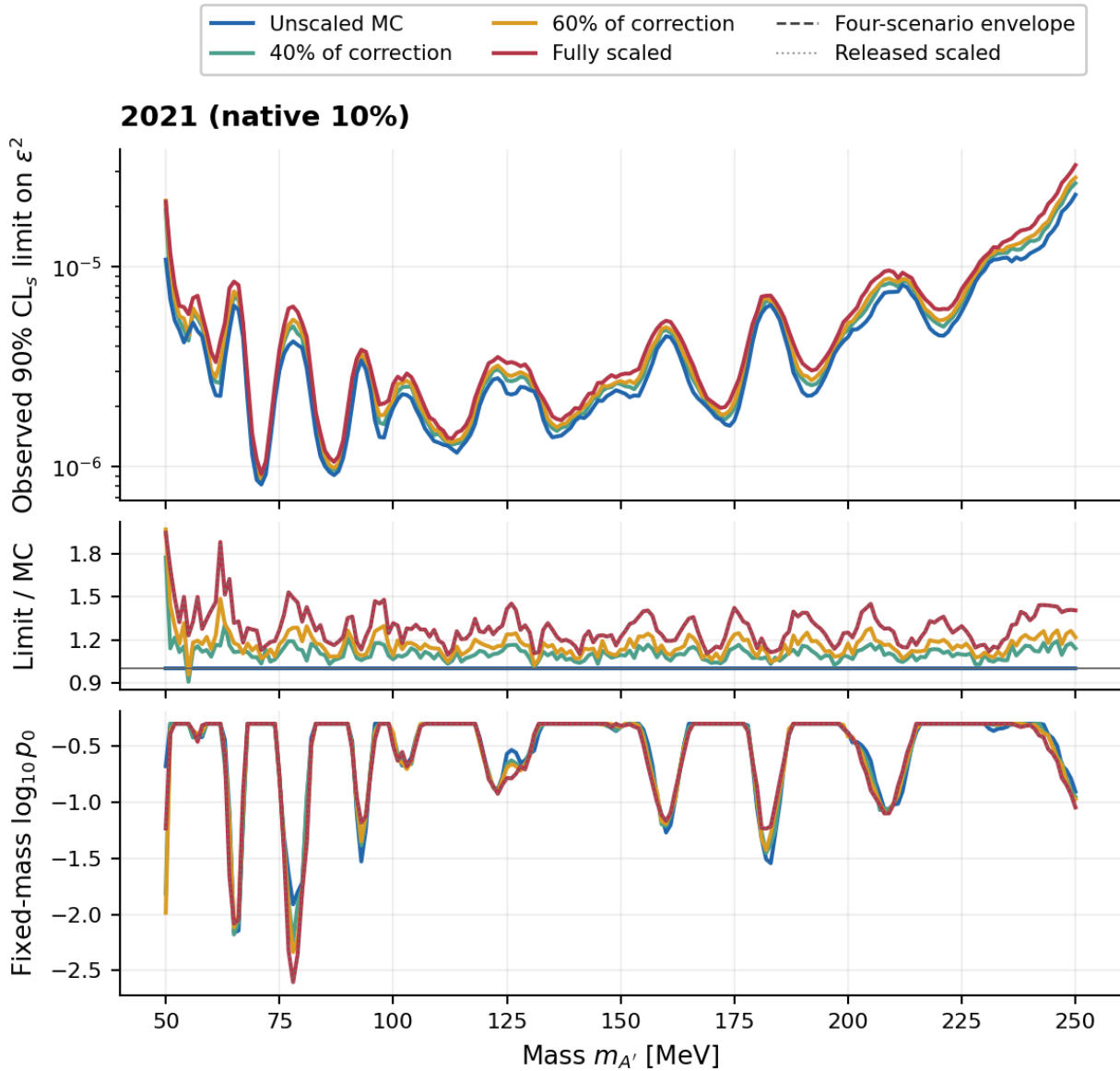


Figure 4. The upper panel compares observed 90% CL_s limits; the middle panel divides each curve by its MC baseline; the lower panel shows fixed-mass $\log_{10} p_0$. The dashed curve is the four-scenario maximum, without a probability interpretation. The gray dotted curve is the frozen scaled release, whose numerical comparison is discussed on page 9.

The fully scaled / MC limit ratio has median 1.27 and spans 1.1-1.95. The maximum four-scenario envelope / MC ratio is 1.97. An intermediate width supplies the largest limit at 1 of 201 masses with all four scenarios.

The smallest MC-scenario p_0 is 0.00686 at 65 MeV; the fully scaled minimum is 0.00249 at 78 MeV. These minima may occur at different masses and are not global p-values.

Combination: one shared coupling

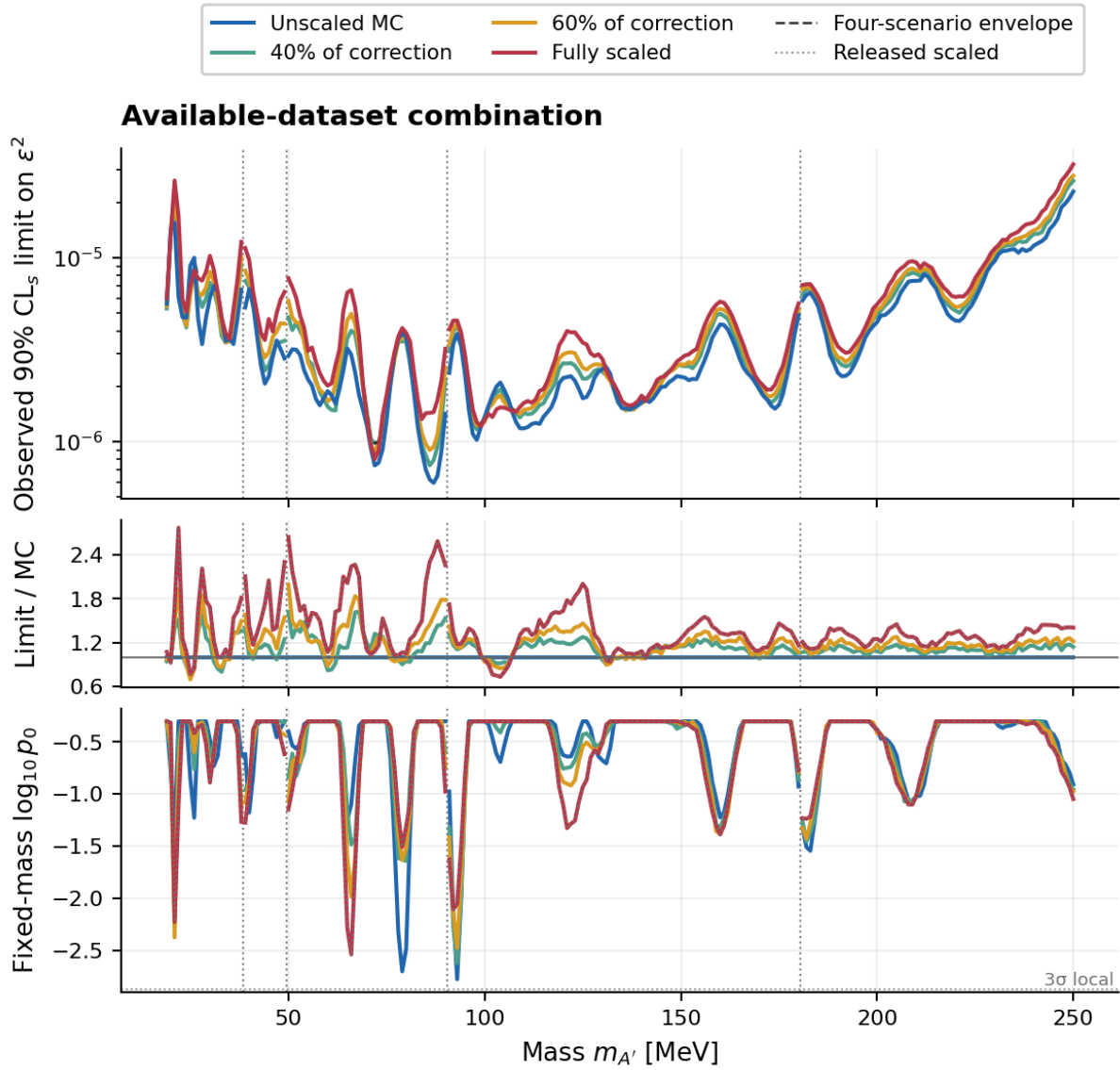


Figure 5. The combination uses each dataset available at a given mass with its own width and count-to-coupling conversion. The common t moves all active datasets along the same progression. Vertical dotted markers indicate a change in dataset membership; line segments are separated at those changes.

Membership: 19-38 MeV: 2015; 39-49 MeV: 2015+2016; 50-90 MeV: 2015+2016+2021; 91-180 MeV: 2016+2021; 181-250 MeV: 2021.

The common progression is a coherent broadening path. It does not establish that the datasets' calibration errors are fully correlated, and its envelope does not sample every independent combination of three width factors.

Minimum local p_0 along each curve: $t = 0$: 0.00167 at 93 MeV; $t = 0.4$: 0.00239 at 93 MeV; $t = 0.6$: 0.00334 at 93 MeV; $t = 1$: 0.00289 at 66 MeV. Selecting among these minima would introduce an additional width-search dimension.

Why the observed curves change

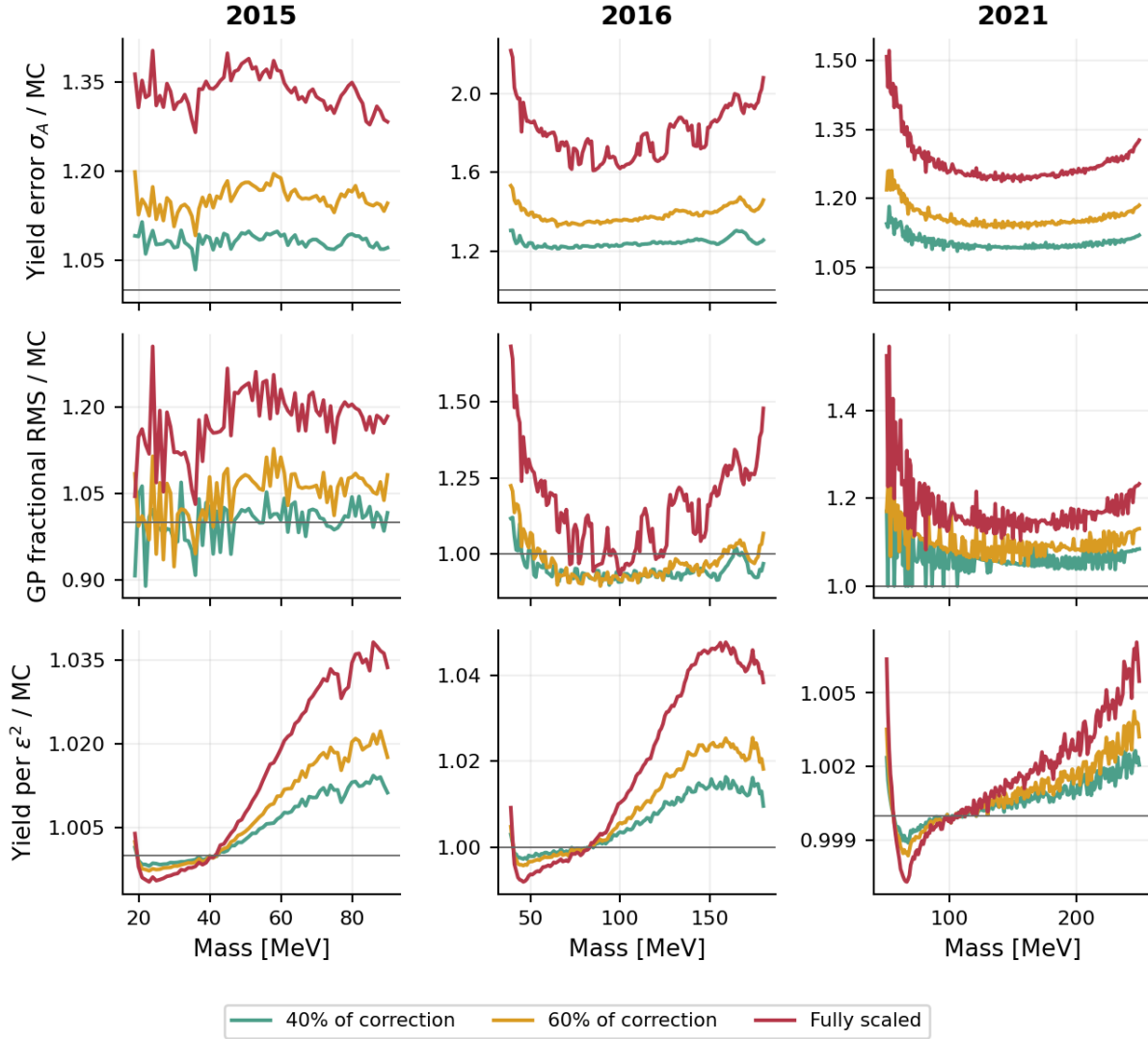


Figure 6. Ratios to the MC scenario of the fitted full-template signal-yield error, the RMS fractional GP prediction uncertainty in the extraction window, and the signal yield per unit ϵ^2 . Changing the width also changes the fitted bins and GP training exclusion; these diagnostics need not vary monotonically.

For a known, locally flat background density β and a normalized Gaussian signal, the Fisher approximation gives $\text{Var}(\hat{A}) = 2\sqrt{\pi} \beta \sigma$. Under those assumptions alone, increasing the width by 25% increases the signal-yield error by $\sqrt{1.25} - 1 = 11.8\%$. This explains why a narrower template can reduce statistical error, but it does not predict an observed upper limit.

In these 2021 scans, the 25% width increase gives a median 26.0% increase in fitted yield error and 27.1% increase in the observed limit. The known-background approximation excludes GP uncertainty and changes to the fit window.

An observed limit also depends on the fitted excess or deficit, GP covariance, signal acceptance within the selected bins, and the prompt-density conversion. An endpoint limit difference cannot be added in quadrature with σ_A ; the former is the nonlinear response of an interval, while the latter describes local fit curvature.

The saved fixed-kernel lane is a conditional control over the same full mass grid. Keeping kernel hyperparameters fixed does not hold the fitted observations, GP conditioning or prompt-density normalization fixed, so it is not a pure signal-shape nuisance profile. Its numerical comparison is included in `derived/report_summary.json`.

What a one-sided uncertainty would require

A defensible calibration treatment needs both the data/MC width ratio and the uncertainty in measuring and transferring that ratio to the signal. An auxiliary calibration likelihood would ordinarily favor the measured correction. A half-normal constraint centered on the MC width, or a hard range from MC to the scaled endpoint, adds a separate modeling assumption that must be justified.

For a genuine resolution nuisance, use a fixed observed bin set and a fixed background construction covering the widest signal, or an explicit joint likelihood including all sideband information. Profile each dataset's width and the background in both numerator and denominator of the likelihood ratio; recompute the corresponding Asimov and CL_s terms. Comparing NLL values from the independently refitted scenarios is not this construction.

The four-point envelope is $U_{\text{env}} = U_{\text{MC}} + \max_t (U_t - U_{\text{MC}}) \geq U_{\text{MC}}$. It answers how far the observed upper limit moves upward on the tested grid. It is not a one-standard-deviation band, and a finite grid need not contain the continuous maximum. The fully scaled endpoint need not give the largest limit.

For discovery, the width is absent from the signal-region model at zero signal. Minimizing p_0 across both mass and width therefore requires a calibrated search procedure; applying the fixed-width asymptotic mapping after that minimization is not justified by these scans. No global discovery probability is reported.

Scope	MC minimum p_0 mass [MeV]	Scaled minimum p_0 mass [MeV]
2015	0.00371 51	0.000847 51
2016	0.00533 93	0.000308 90
2021	0.00686 65	0.00249 78
Combination	0.00167 93	0.00289 66

Statistical sources. PDG, [Statistics \(2025\)](#), [auxiliary measurements](#); [Cowan et al., arXiv:1007.1727](#), profile likelihood and asymptotic tests; [Gross and Vitells, arXiv:1005.1891](#), search parameters present only under the alternative.

Reproducibility. This standalone folder contains protocol.json, derived/scans.csv, derived/report_summary.json, resolution_audit.md, statistics_review.md, the source snapshots and plotting/fit scripts. The plot builder reads the completed scan rows and performs no new statistical fits. Numerical convergence checks and source identities are recorded in the study's QA and provenance files.

Source choices and numerical controls

2015. The MC curve is inferred by dividing the production curve by $1.03 \times (1.609/1.303)$, removing the target systematic factor and rounded Møller correction. It is not a newly fitted raw-MC curve. **2016.** The raw and track-smeared Table 12 polynomials are interpolated pointwise; the full / MC ratio spans 1.811-2.071 over the scan. The source notes a low-mass polynomial deviation below 40 MeV; the retained 39 MeV edge inherits that caveat. The 180 MeV upper edge avoids tail extrapolation. **2021.** The source MC already includes Omega and hit smearing; unscaled removes only the additional target-constrained Møller factor. The exact production factor 1.25 is retained.

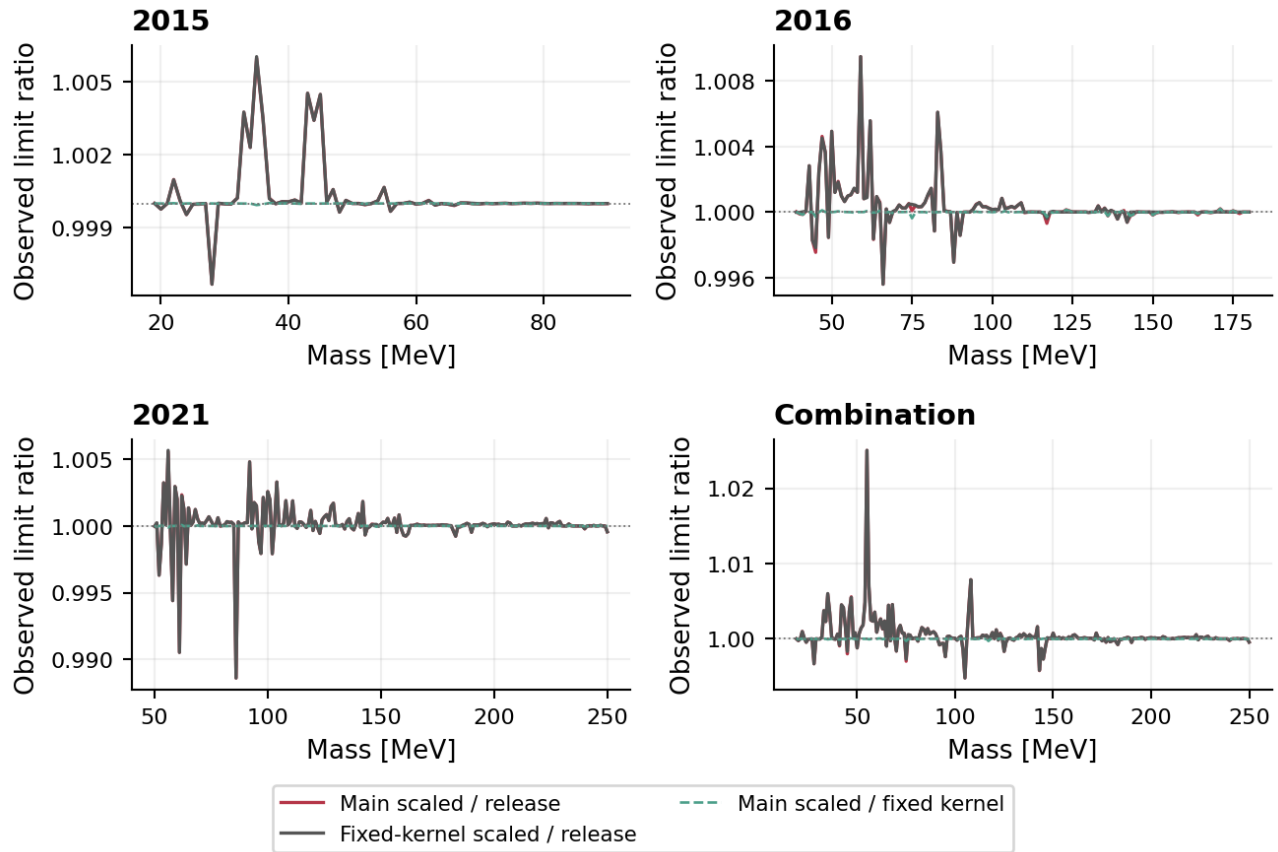


Figure 7. At fully scaled width, the red and gray curves divide the main result and fixed-kernel control by the frozen release. The green dashed curve divides the main result by the fixed-kernel control. The common width makes these procedural comparisons, rather than resolution variations.

The solver evaluates Poisson deviance and checks profile score residuals, reducing cancellation from subtracting large absolute log likelihoods. Differences from the frozen solver can therefore remain even with reviewed kernels held fixed. Reoptimization can introduce a further change. These controls are numerical and procedural checks; they are not uncertainty bands.

At identical fully scaled widths, the largest absolute relative limit difference from the frozen release is 2.50% in the fixed-kernel control and 2.5% after GP reoptimization (combined, 55 MeV). That latter difference is procedural, not an effect of changing the resolution.

Saved scan checks: 7764 rows; largest CL_s residual $1.39e-08$; largest profile score residual $2e-07$. The machine-readable validation record retains any failed or recovered cells.